Patient-robust discovery of combinatorial cell-surface targets in prostate cancer

AND-gate and NOT-gate marker pairs from single-cell data, benchmarked against PSMA-PSCA

William Guesdon

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Abstract Single-antigen therapies against solid tumors are limited by on-target off-tumor toxicity, because few surface antigens are truly absent from all healthy tissue. Combinatorial (logic-gated) targeting requires two conditions before a cell is engaged, which can recover specificity. We scan a curated cell-surface panel in single-cell data for pairs that discriminate prostate cancer cells from healthy human cell types, under AND (both markers on the tumor) and NOT (an activator spared by a blocker on healthy cells) logic. Pairs are scored per patient and summarized across patients, so no high cell-count patient drives a result. The clinically validated PSMA-PSCA pair is recovered as a positive control: each marker alone carries a large extra-prostatic liability, and the AND gate collapses it. We report a Pareto frontier of surface-accessible pairs that improve per-patient coverage over PSMA-PSCA at comparable off-tissue risk, with Human Protein Atlas evidence qualifying targetability.

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1 Background

Most targeted therapies against solid tumors fail on safety, not potency. A single tumor antigen is rarely absent from every healthy tissue, so single-target CAR-T, T-cell engagers, and antibody-drug conjugates carry on-target off-tumor toxicity. Combinatorial, logic-gated targeting requires two conditions before a cell is engaged and can recover the missing specificity [1]–[3]. Prostate adenocarcinoma is the natural test bed: the PSMA (FOLH1) x PSCA pair is a preclinically validated AND-gate split-signal CAR [1], so an unsupervised scan can be checked against a known answer, and STEAP1 anchors a second axis with a STEAP1 x CD3 T-cell engager (xaluritamig) in trials [4].

Computational logic-gated antigen discovery is not itself new: pan-cancer single-cell pair scans [5] and a breast-cancer AND/OR/NOT circuit search [6] precede this work. Our contribution is the combination of (i) per-patient scoring rather than pooled cells, (ii) a matched benign-prostate control that tests tumor specificity in the same samples, and (iii) recovery of the PSMA-PSCA benchmark before any new pair is nominated.

2 Data

- **Tumor discovery cohort** — 68,322 cells from **24 patients** with localised, hormone therapy-naive prostate adenocarcinoma [7]. Malignant identity uses the authors' copy-number and signature-based annotation: **3,590 malignant cells** (scDblFinder singlets; see Methods), with **16,249 adjacent-benign / normal prostate epithelial cells** retained as a matched same-patient control.
- **Healthy reference** — Tabula Sapiens [8], pulled through the CELLxGENE Census with a server-side subset to the panel genes: **1,136,218 cells**, **180 cell types** across **75 tissues** (585 tissue-by-cell-type populations scored). This is the safety denominator.
- **Protein evidence** — Human Protein Atlas subcellular localization and normal-tissue RNA [9], used to qualify surface accessibility and cross-check normal-tissue liabilities.
- **Panel** — 29 curated cell-surface markers scanned (see Methods); secreted markers (KLK3/PSA, KLK2, ACPP) are kept only as comparators.

3 Methods

- **Doublet removal (primary input).** scDblFinder, run per patient in a Bioconductor container, flagged **4.8%** of tumor cells as doublets (5.6% of malignant cells); these are removed before scoring, so a two-cells-as-one capture cannot fake AND co-expression (Rule 8). Re-scoring on the 65,040 retained single cells moves the key pairs' AND coverage by at most 1.0 percentage points, so the finding is not a doublet artifact.

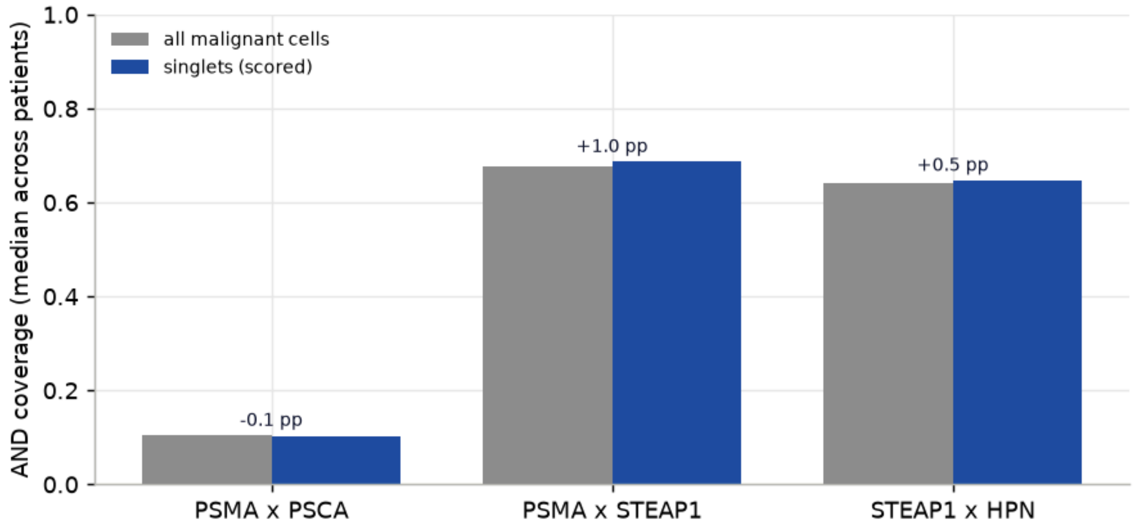

Figure 1: Doublet robustness. Median per-patient AND coverage of the key pairs computed on all malignant cells (grey) versus the scDblFinder singlets actually used for scoring (blue). The bars are near-identical and the change (annotated, percentage points) is at most one point, so the co-expression signal is not an artifact of two cells captured as one; removing doublets nudges the nominated pairs slightly upward.

- **Positivity** is a raw count of at least 1 per cell (detection), never a single `count > 0` shortcut baked into the summary; the threshold is swept for sensitivity (below).
- **Per-patient scoring (never pooled).** For markers A, B and each patient's malignant cells, AND coverage is `P(A+ and B+ | malignant)` computed within that patient, then summarized across patients by the lower decile (Q0.10, the robustness floor) and the median. A patient with thousands of cells cannot dominate a pair's score.
- **Three references per pair.** (1) malignant coverage; (2) **matched benign prostate** from the same cohort, so a pair expressed across all prostate cells is penalised, not just one specific to the tumor;

  (3) **healthy liability** as the worst-case co-positive fraction over Tabula Sapiens populations, reported both across all organs and **excluding the prostate** (the target organ is dispensable; every other organ is the dose-limiting risk).
- **Gate algebra.** For each group a single matrix product over the panel positivity matrix yields AND, NOT (`A+ and B-`) and OR counts for every pair at once.
- **Ranking.** Pairs are placed on a Pareto frontier of per-patient coverage floor (higher better) against worst extra-prostatic liability (lower better). Only surface-accessible markers are nominatable.
- **Reproducibility.** Every number below is read from a committed table in `results/tables/`; the report redraws its figures from those tables with no network or model call.

4 Results

4.1 Single surface antigens are not safe alone

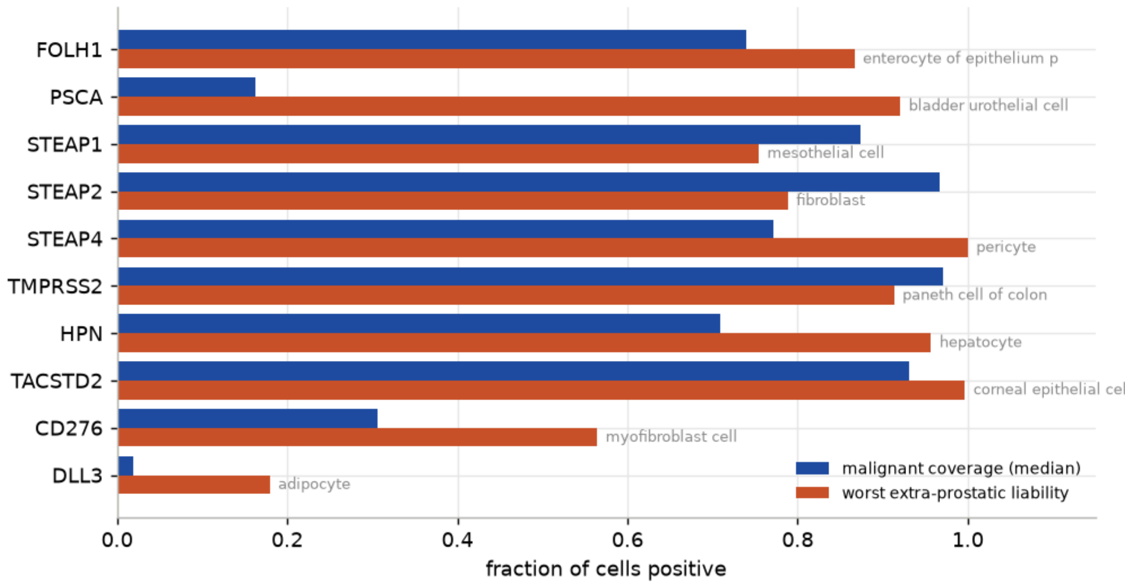

Figure 2: Single surface antigens: per-patient malignant coverage (median) against the worst-case extra-prostatic healthy liability. Every strong single antigen carries a large off-tissue liability; the recovered worst tissues match known biology (FOLH1/PSMA to duodenum, PSCA to bladder urothelium).

- Each strong single antigen is broadly positive on the tumor but also on at least one healthy cell type: the scan independently recovers PSMA's duodenal and PSCA's bladder-urothelial expression, so the liability calls are trustworthy.

4.2 The AND gate recovers PSMA-PSCA and collapses its single-agent liabilities

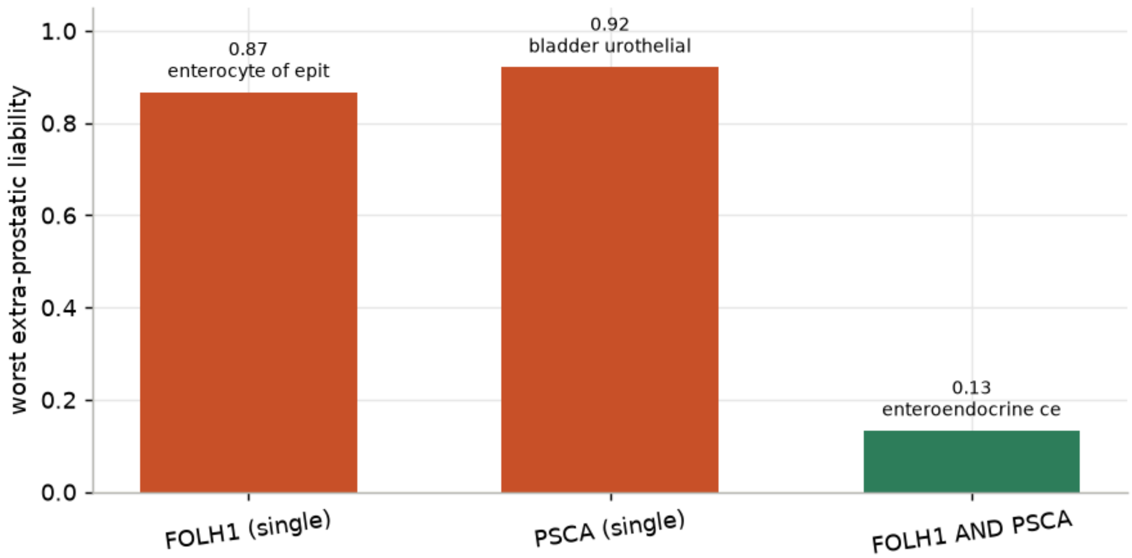

Figure 3: Positive control. PSMA (FOLH1) and PSCA are each broadly expressed off the prostate (duodenum, bladder), but requiring both at once (AND) collapses the worst extra-prostatic liability roughly sevenfold, recovering the clinical rationale for the split-signal PSMA x PSCA CAR.

- **Recovered.** PSMA alone reaches **0.87** in duodenal enterocytes and PSCA alone **0.92** in bladder urothelium; the **PSMA and PSCA** gate drops the worst extra-prostatic liability to **0.13** (a reduction of 0.79).
- **Above the random baseline (Rule 7).** 80% of random surface pairs score worse than PSMA-PSCA on extra-prostatic selectivity, and PSMA-PSCA sits on the surface-accessible Pareto frontier.
- Its low raw coverage (0.10 median) reflects PSCA transcript dropout, not a biology failure; this is the motivation to look for higher-coverage pairs of equal safety.

4.3 A Pareto frontier of surface pairs improves on PSMA-PSCA

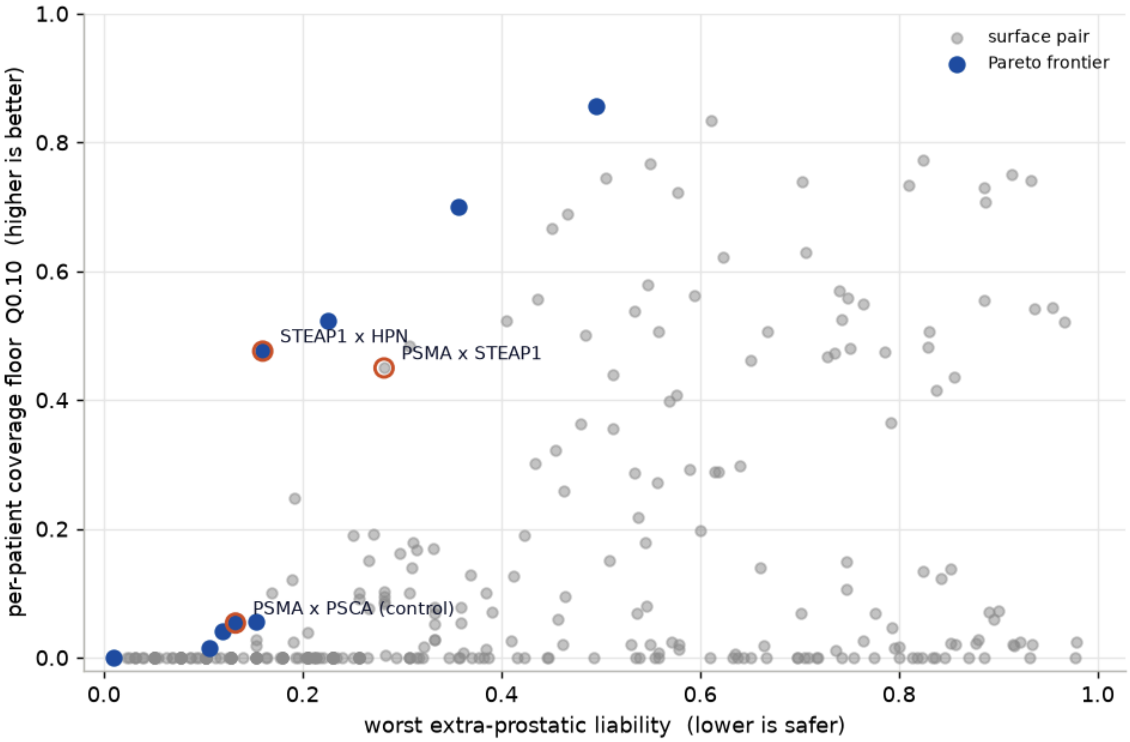

Figure 4: Surface-accessible AND pairs: per-patient coverage floor (Q0.10) against worst extra-prostatic liability. Upper-left is better. Frontier pairs are filled; the PSMA-PSCA positive control and the two co-lead nominations are labelled. Secreted markers and intracellular prostein are excluded as non-targetable.

Two surface-accessible co-leads improve on the control (both recovered as safe, higher-coverage pairs):

- **PSMA x STEAP1 — translatable lead.** Both antigens have clinical-grade binders. Per-patient coverage floor **Q0.10** = **0.45**, median **0.69** (**6.7x** the PSMA-PSCA median), worst extra-prostatic liability **0.28** in adipocyte, versus 0.87 / 0.75 for the single antigens.
- **STEAP1 x HPN — Pareto-optimal lead.** Coverage floor **Q0.10** = **0.48**, median **0.65**, worst extra-prostatic liability **0.16** (mesothelial cell) — the lowest off-tissue risk among high-coverage pairs.
- **Tumor-specific, not pan-prostate.** Both leads separate malignant from matched benign prostate (malignant-minus-benign coverage 0.60 for PSMA x STEAP1), so the gate marks cancer rather than the whole gland.

4.4 Robustness: per patient and across positivity thresholds

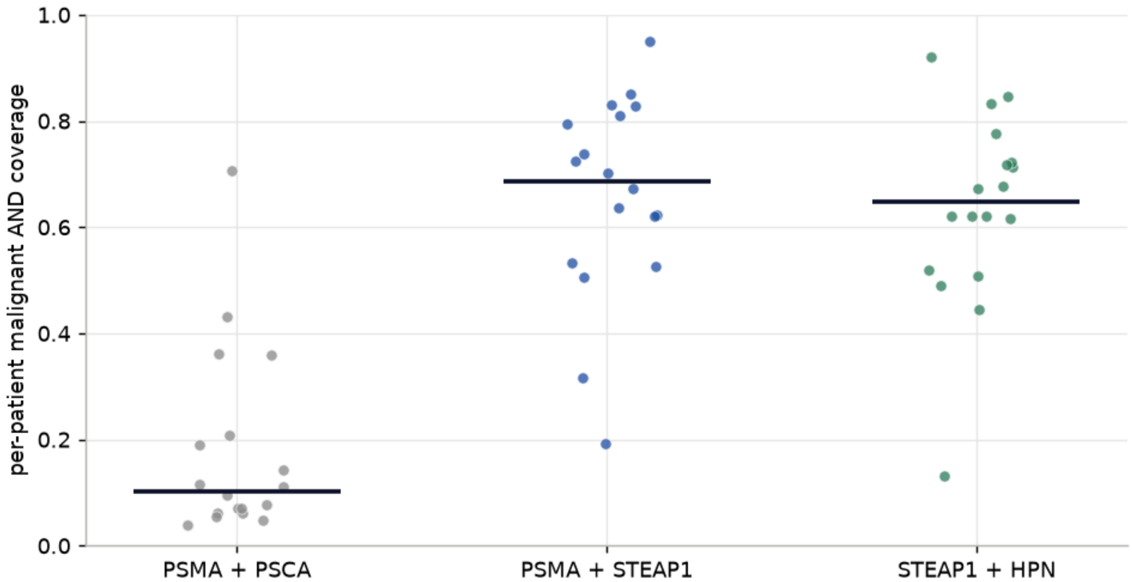

Figure 5: Per-patient AND coverage of the malignant compartment for the positive control and the two co-lead nominations. Each point is one patient; the bar is the median. PSMA x STEAP1 and STEAP1 x HPN cover the tumor in every scored patient, while PSMA x PSCA is uniformly low (PSCA transcript dropout).

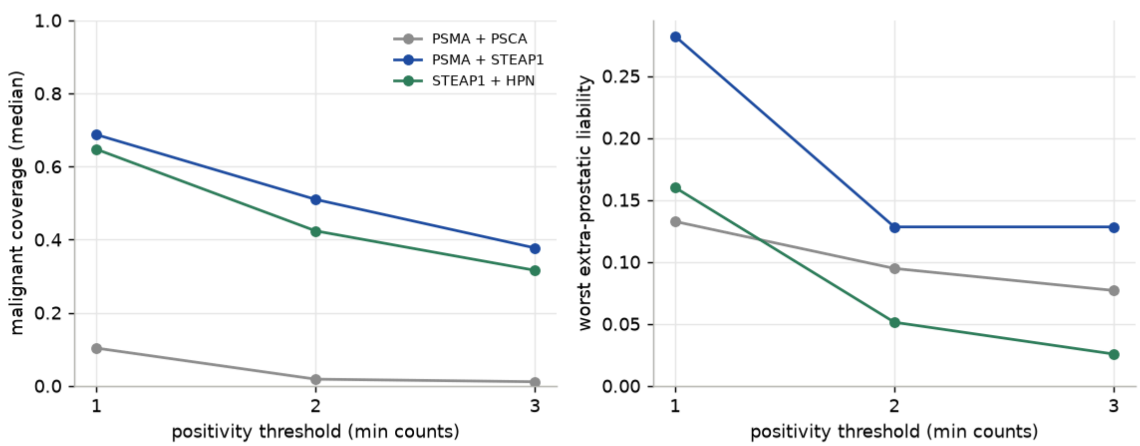

Figure 6: Positivity-threshold sensitivity. As the per-cell call is tightened from >=1 to >=3 counts, coverage falls modestly but the worst extra-prostatic liability also falls: the nominated pairs stay well covered and become safer, whereas PSMA x PSCA loses coverage. Positivity is never a single count>0 call.

- Both nominations survive threshold tightening: coverage decreases gradually while the worst extra-prostatic liability falls, so the safety claim is not an artifact of the loosest positivity call.

Residual off-tumor liabilities of PSMA x STEAP1 are named, not hidden. The highest extra-prostatic healthy populations (AND co-positive fraction):

tissue	cell_type	coverage	n_cells
eye	adipocyte	0.282	39
small intestine	paneth cell of epithelium of small intestine	0.136	1853
respiratory system	mucus secreting cell	0.119	270
tongue	salivary gland cell	0.096	848
small intestine	intestinal crypt stem cell of small intestine	0.09	587
lymph node	stromal cell	0.081	111

4.5 Exploratory NOT gates

NOT gates (activator on the tumor, blocker sparing a normal tissue) are reported as exploratory: the blocker-negative call depends on trusting scRNA dropout. Top surface activator / normal-tissue blocker candidates:

activator	blocker	tumor_q10	tumor_median	worst_healthy_x-prostate	worst_xp_celltype
STEAP1	MUC1	0.543	0.732	0.509	mesothelial cell
STEAP2	SLC34A1	0.936	0.968	0.789	fibroblast
STEAP2	CDH16	0.936	0.968	0.789	fibroblast
STEAP2	MUC1	0.6	0.843	0.696	fibroblast
STEAP1	SLC34A1	0.739	0.874	0.754	mesothelial cell

5 Limitations

- **Transcript is not targetability.** scRNA measures mRNA, not surface antigen density, epitope accessibility, shedding, or internalization. HPA localization is used to exclude secreted markers and intracellular prostein, but same-cell surface co-expression needs CITE-seq, multiplex IHC, or dual flow to confirm.
- **NOT-gate calls trust B-negatives**, which are dropout-sensitive; they are exploratory.
- **One cohort, localised disease.** The discovery cohort is hormone-naive localised prostate cancer; replication in an independent and a castration-resistant / neuroendocrine cohort is future work.
- **Malignant labels are the authors' CNV/signature calls**, adopted rather than re-derived here.
- **PSCA transcript dropout** caps PSMA-PSCA coverage; the pair is validated on safety, not coverage.

6 Reproducibility

The pipeline runs from a clean clone with `uv`; every table in `results/tables/` is regenerated by the numbered scripts in `scripts/`, and this report redraws its figures from those tables. See the README for the exact command sequence.

7 References

Bibliography

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